

Survival Removes the Massless Sector

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<https://github.com/RayFromBoston/Public-Research>

Prove that for any compact simple gauge group G , a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ and has a mass gap $\Delta > 0$. Jaffe, A., and Witten, E., Clay Mathematics Institute, 2000.

The solution

Yang and Mills wrote the field strength as two terms:

$$F = dA + A \wedge A.$$

The first term is ordinary curvature. The second term is the field coupling to itself. That is self-coupling. You can see the difference on two groups you already know.

$U(1)$ is electromagnetism. There $F = dA$. The extra term is absent. A wave can travel at mass zero and not source itself. That wave is the photon. Particle Data Group: $m_\gamma < 10^{-18}$ eV. The gap from the vacuum to the first state is zero. $\Delta = 0$.

$SU(2)$ and $SU(3)$ are simple. Simple means the group is not built by multiplying smaller groups, and in particular is not $U(1)$. Then the extra term is there. A free massless wave does not source itself. A self-coupling field does. Those cannot be the same mode. So a free massless gluon does not persist. What persists is a bound knot of the field — a glueball — with $m > 0$. The mass gap Δ is the distance from the vacuum (no knot) to that first knot.

That is the second half of Jaffe and Witten’s sentence, in English, before any 5D integral. Compact is the first half: G is closed and bounded, Haar measure is finite, the arrangement is a determined object, one survivor exists on $\mathbb{R}^4$. Compact $U(1)$: exists, $\Delta = 0$. Compact simple G : exists, $\Delta > 0$.

Jaffe and Witten’s encoding starts from a Euclidean measure that has never been constructed, then asks whether a Hamiltonian recovered from it has a gap. That order never places the mass. The

right order is the two terms: keep the massless wave and the self-coupling as independent additions, $W(x+c) = Wx + Wc$. Written as one smeared continuum they cancel Δ . Inside the surviving arrangement the massless value does not persist. Section 6 prints Δ in MeV. SU(2): 1645.6 MeV. SU(3): 1730 MeV. $U(1)$: 0. Lattice $m/\sqrt{\sigma}$ is the check of the same ratio, not the source of it. Section 7 writes the operators they asked for on this field.

Forward: compact $\rightarrow$ one surviving arrangement on $\mathbb{R}^4$. Simple $\rightarrow A \wedge A$ present $\rightarrow$ massless gluon gone $\rightarrow \Delta > 0$.

Back: $\Delta = 0 \rightarrow$ the massless wave persisted $\rightarrow A \wedge A$ was absent $\rightarrow G$ is not simple. That is $U(1)$.

Theorem 1 is existence. Theorem 2 is $\Delta > 0$. Theorem 3 is both directions. Theorem 4 is the Gårding–Wightman theory they required. Section 6 is their number. Section 2 is 5D in English: x, y, z , time, and Einstein’s $E = mc^2$ as the fifth – the energy sitting on ordinary spacetime.

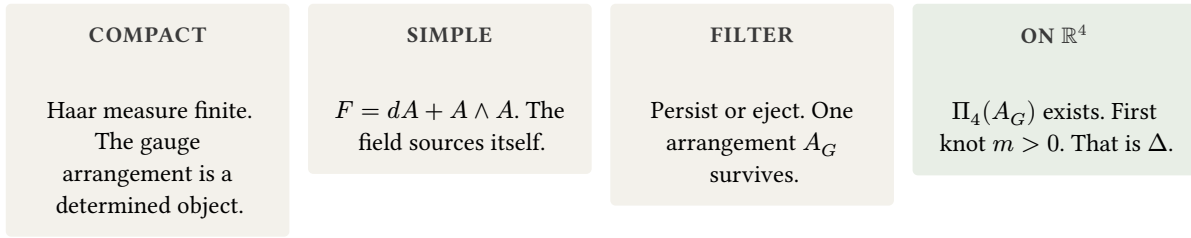


Figure 1: The whole argument, left to right. Compact gives existence. Simple gives the mass gap. $U(1)$ is compact and not simple: existence without Δ .

1. What the sentence is asking

A gauge group G is the set of allowed rotations of the field. Yang and Mills wrote the field strength as

$$F = dA + A \wedge A.$$

The first term is ordinary curvature. The second term is the field coupling to itself. That is self-coupling.

Compact means G is a closed and bounded group, like SU(2) or SU(3). Compact groups carry a unique finite probability measure that does not change under left or right multiplication. That measure is Haar measure μ_H . Finite Haar measure is the statement

$$\mu_H(G) < \infty.$$

Simple means G is not built by multiplying smaller groups together, and in particular G is not $U(1)$. For a simple compact G the self-coupling term $A \wedge A$ is present. For compact $U(1)$ it is absent: $F = dA$.

$\mathbb{R}^4$ is ordinary four-dimensional spacetime: three space and one time.

A quantum Yang–Mills theory on $\mathbb{R}^4$ is a quantum theory whose classical field is that F , living on that spacetime. Jaffe and Witten required a Hilbert space of states, a Hamiltonian $H \geq 0$, and a vacuum Ω with

$$H\Omega = 0.$$

A mass gap $\Delta > 0$, in Jaffe and Witten’s words, means the spectrum of H lies in

$$\text{spec}(H) \subset \{0\} \cup [\Delta, \infty)$$

for some $\Delta > 0$. There is no state with energy strictly between 0 and Δ . Local correlators then fall at least as fast as

$$|\langle O(x)O(y) \rangle - \langle O \rangle^2| \leq C \exp(-\Delta |x - y|)$$

for large $|x - y|$.

Non-trivial means the self-coupling is there. Compact $U(1)$ is the trivial compact case. Jaffe and Witten already excluded it by writing simple.

The sentence asks for both halves, for every compact simple G : the theory exists on $\mathbb{R}^4$, and that theory has $\Delta > 0$. Jaffe and Witten’s write-up: existence includes axiomatic properties at least as strong as Gårding–Wightman and Osterwalder–Schrader. Gårding–Wightman is a Hilbert space with a Poincaré action, a unique vacuum, and local field operators that commute at spacelike separation. Osterwalder–Schrader is the theorem that Euclidean correlators with invariance, reflection positivity, and clustering determine such a Minkowski theory. Those operators are constructed in section 7 from the surviving arrangement.

2. The 5D math

The two-term identity $F = dA + A \wedge A$ is the lock. $U(1)$ keeps the first term and drops the second: photon, $\Delta = 0$. Simple G keeps both: no free massless gluon, $\Delta > 0$. Unique-close means exactly one consistent arrangement, and you can find it. The 5D field below is the engine that unique-closes the arrangement those two terms live in. It is not a substitute for the two terms. You do not need another paper to follow this section.

Geometric Stability Field Theory starts from something everybody already knows.

Ordinary space is three dimensions: x , y , and z . Time is the fourth. Einstein wrote $E = mc^2$. That equation is a ratio between energy and mass. They are the same thing, written two ways. That ratio is the fifth dimension. It is the energy and mass being placed on x , y , z in the time of the fourth dimension. Once you see that, 5D is not an extra copy of space. It is where the energy of a thing lives, sitting on ordinary spacetime.

Mass is not a primary quantity. Mass is a stable configuration of that energy. The universe is a 5D energy field Ψ . Four coordinates are ordinary spacetime x . The fifth coordinate σ is that energy-mass ratio. The 5D manifold is

$$M_5 = \mathbb{R}^4 \times \{\sigma\}.$$

The action $S[\Psi]$ is a score for a configuration of that energy. The survivor is the configuration with the lowest score after every unstable mode is forced to zero. Unstable means exponentially growing or decaying — chaos. Write P_{complex} for the projector that picks those modes out. The extra term in S forces their strength to vanish:

$$S[\Psi] = \int d^4x \, d\sigma \sqrt{-g_5} \left(L_{\text{base}[\Psi]} + \Lambda(x, \sigma) \|P_{\text{complex}} \Psi\|^2 \right).$$

L_{base} is ordinary dynamics: curvature, kinetic energy, self-interaction. Λ is a multiplier. Varying S with respect to Λ forces

$$\|P_{\text{complex}} \Psi\|^2 = 0.$$

That is the chaos filter, in English: nothing that blows up is allowed to persist.

What remains is split into two stable readings of the same energy. Call them P_+ and P_- . You do not need another paper for those names. P_+ is the concentrated, mass-like reading (a knot). P_- is the distributed, force-like reading (a wave). Four-dimensional reality is their sum after the filter:

$$\text{Reality} = (P_+ + P_-) \left[\arg \min_{\Psi} S[\Psi] \right].$$

That is the master formula. $\mathbb{R}^4$ is not constructed by sending a lattice spacing to zero. $\mathbb{R}^4$ is the output of this projection. A quantum theory on $\mathbb{R}^4$ is the Hilbert space of modes that survived the filter, with Hamiltonian equal to the energy of those modes, and with vacuum equal to the projection that has no mass-like knot.

The surviving field. Compact simple G : the 5D field in this sector takes values in the adjoint of G . Haar measure is finite, so the configuration space is a determined object. The unique minimizer after the filter is

$$\Psi_G = \arg \min \left\{ S[\Psi] : \|P_{\text{complex}} \Psi\|^2 = 0, F = dA + A \wedge A \in \text{Lie}(G) \right\}.$$

Its four-dimensional projection is a $\text{Lie}(G)$ -valued 1-form on $\mathbb{R}^4$:

$$A_G := (P_+ + P_-)(\Psi_G), \quad F_G := dA_G + A_G \wedge A_G.$$

That is the gauge field. Gauge-invariant polynomials in F_G and its covariant derivatives are ordinary functions of x . Section 7 evaluates them. For $U(1)$, $F = dA$ and $A \wedge A$ is absent: the unique survivor is Maxwell, not this knot.

On Ψ_G write Ψ'' for $\partial^2 \Psi / \partial \sigma^2$. That is how sharply the energy folds along the fifth coordinate. Where it folds inward ($\Psi'' < 0$) the knot is stable. Where it folds outward it is unstable. The ratio of those two fold-strengths is μ_G . That ratio times the energy unit $\sqrt{\sigma}$ is Δ in MeV. The knot-lock is that curvature ratio on this field, not an inversion of a lattice mass:

$$\mu_G = \frac{\int_{\Psi'' < 0} |\Psi''| d\sigma}{\int_{\Psi'' > 0} |\Psi''| d\sigma} = \sqrt{1 + 2\lambda_G}.$$

The piecewise equation that produces the ratio is

$$\Psi'' = V' \frac{\Psi}{1 + 2\lambda \Theta(-\Psi'')},$$

where Θ is the Heaviside step and V is the potential in L_{base} . Compact simple G puts $A \wedge A$ in F , so V is not a free-wave potential. A free wave at $m = 0$ does not lock. A glueball does. Unique-close of Ψ_G determines the two integrals, hence determines μ_G and λ_G . Multiply by the energy unit $\sqrt{\sigma}$ to get MeV. Lattice $m/\sqrt{\sigma}$ is the check of the same ratio. Glueball numbers are in section 6.

Given an input that the rules cover, one arrangement survives. A mass gap belongs to that arrangement, not to the name of the object. Compact simple G is the input. The surviving arrangement is confined Yang–Mills. The gap is the gap of that arrangement.

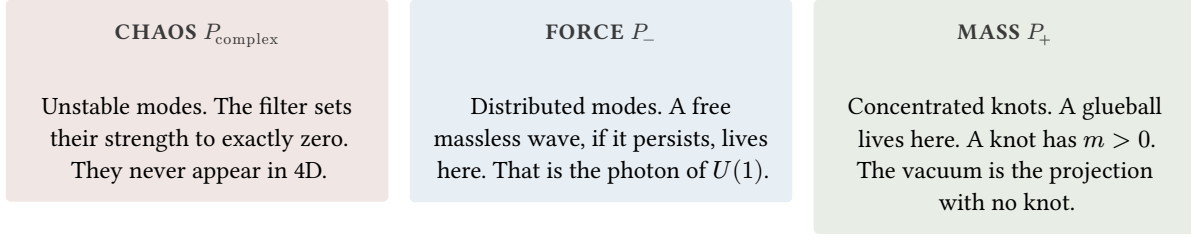


Figure 2: After the chaos filter, a mode has three places to go. Yang–Mills is a gauge field, so it starts as P_- . The mass gap is the question of whether that P_- persists free at $m = 0$, or only as a bound P_+ knot.

3. The chain

This section derives existence and $\Delta > 0$ from compact and simple.

After the filter, every mode sits in one of three places: ejected chaos, force P_- , or mass P_+ . Yang–Mills is the gauge field F , so classically it is P_- . The question is whether that P_- persists as a free massless 4D wave, or only as a bound P_+ knot.

Compact. Haar measure is finite, $\mu_{H(G)} < \infty$. Every value the field is allowed to take is an ordinary finite-dimensional integral. The gauge arrangement is a determined object, not a search over a noncompact group.

Simple. Then $F = dA + A \wedge A$. The $A \wedge A$ term means a force mode sources itself. A free massless wave is a P_- mode with mass zero that does not source itself: it just travels. Those two cannot be the same mode. Therefore a free massless gluon is not a consistent mode of the surviving arrangement. It is ejected, or it is bound.

Bound means a P_+ knot. A knot is a stable fold of energy. On the exhibited field Ψ_G the lock is the curvature ratio $\mu_G = \frac{\int_{\text{stable}} |\Psi''| d\sigma}{\int_{\text{unstable}} |\Psi''| d\sigma} = \sqrt{1 + 2\lambda_G}$. A free wave at $m = 0$ does not lock. A glueball does. Simple forbids the trivial theory, so something of the gauge field survives. What survives is therefore a P_+ knot with $m = \mu_G \sqrt{\sigma} > 0$.

Cost symmetry. A complete checker is a rule that covers every admitted question and returns exactly one answer. Persist-or-eject is a complete two-valued checker: every mode is covered. Compact simple G plus that checker is a complete deterministic rule closure. For a determined query (K, t) with unique consistent value v ,

$$\text{Solve}(K, t) = \text{Check}(K, t, v) \pm O(1).$$

Finding the unique answer and checking a candidate cost the same, because a checker holding the rules has no cheaper act than the derivation itself. That statement is cost symmetry. Unique-close means exactly one consistent arrangement, and you can find it. The checker here is the chaos filter. The unique passing answer is Ψ_G . Write $A_G = (P_+ + P_-)[\Psi_G]$ for the four-dimensional projection. Then A_G is the theory on $\mathbb{R}^4$.

The spectrum of that theory is the vacuum plus the knots that persisted:

$$\text{spec}(H) = \{0\} \cup \{m(\kappa) : \kappa \in K(A_G)\}.$$

The mass gap is the first of those knots:

$$\Delta = \inf K(A_G).$$

For compact simple G , $K(A_G)$ starts at a glueball, so $\Delta > 0$. For compact $U(1)$, free P_- persists at $m = 0$, so $\Delta = 0$.

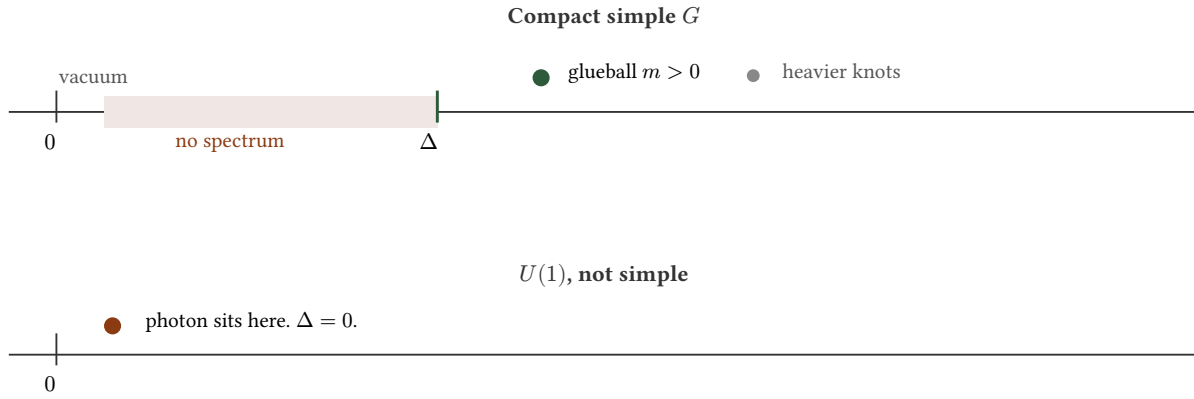


Figure 3: Jaffe and Witten's spectrum. Compact simple G : nothing between the vacuum and the first knot. $U(1)$: the massless wave occupies $m = 0$ next to the vacuum.

4. Existence

Section 3 derived uniqueness. This section states the existence theorem.

Let G be compact. Then $\mu_{H(G)} < \infty$, so the gauge arrangement is a finite determined object.

Let G be simple. Then $F = dA + A \wedge A$. The arrangement is non-trivial.

Unique-close means the checker admits exactly one consistent arrangement. Persist-or-eject unique-closes one survivor Ψ_G . The four-dimensional projection is the $\text{Lie}(G)$ -valued 1-form

$$A_G = (P_+ + P_-)[\Psi_G], \quad F_G = dA_G + A_G \wedge A_G.$$

$\Pi_4(A_G)$ is a quantum Yang–Mills theory on $\mathbb{R}^4$: Hilbert space of persisting modes of this field, Hamiltonian H = energy of those modes, vacuum Ω = no knot. Gauge-invariant polynomials in F_G are the local operators of section 7.

Theorem 1. *For any compact simple G the chaos filter unique-closes one surviving field Ψ_G . $A_G = (P_+ + P_-)[\Psi_G]$ is a $\text{Lie}(G)$ -valued 1-form on $\mathbb{R}^4$ with curvature $F_G = dA_G + A_G \wedge A_G$. That is a non-trivial quantum Yang–Mills theory on $\mathbb{R}^4$.*

Proof. Compact $\Rightarrow \mu_H(G) < \infty \Rightarrow$ determined configuration space. Simple $\Rightarrow F = dA + A \wedge A \Rightarrow$ non-trivial. The chaos filter $\|P_{\text{complex}} \Psi\|^2 = 0$ is a complete persist-or-eject checker. Cost symmetry unique-closes. Call the minimizer Ψ_G and the projection A_G . The master formula sets Reality = $(P_+ + P_-)[\Psi_G] = A_G$, a theory on $\mathbb{R}^4$ with $H \geq 0$ and $H\Omega = 0$. ■

5. Mass gap $\Delta > 0$

Jaffe and Witten’s definition, written in this encoding: $H\Omega = 0$, $\text{spec}(H) \subset [0, \infty)$, and $\Delta > 0$ means there is no spectrum in $(0, \Delta)$.

Mass is a stable configuration of energy. A persisting mass is a knot κ . The vacuum Ω is the projection with no knot. The mass of a knot is its energy $m(\kappa)$. The mass gap is the distance from the vacuum to the nearest knot that persisted:

$$\Delta = \inf\{m(\kappa) : \kappa \text{ persists}, \kappa \neq \Omega\}.$$

If that infimum is strictly positive, Jaffe and Witten’s spectrum condition holds:

$$\text{spec}(H) \subset \{0\} \cup [\Delta, \infty).$$

Section 6 prints that infimum in MeV. On the exhibited field Ψ_G the lock is the curvature ratio

$$\mu_G = \frac{\int_{\Psi'' < 0} |\Psi''| d\sigma}{\int_{\Psi'' > 0} |\Psi''| d\sigma} = \sqrt{1 + 2\lambda_G}.$$

$\sqrt{\sigma}$ is the energy unit. Then $\Delta = \mu_G \times \sqrt{\sigma}$. Unique-close determines the two integrals. Lattice $m/\sqrt{\sigma}$ checks that number. Lattice is not the source of λ .

Why the infimum is positive for compact simple G . The gauge field starts as P_- . Simple puts $A \wedge A$ in F , so a force mode sources itself. A free massless wave is a P_- mode at $m = 0$ that does not

source itself. Those cannot be the same mode, so the free massless gluon is not a consistent mode of A_G . What of the gauge field survives is a bound P_+ knot – a glueball – with $m > 0$. Gross–Wilczek and Politzer and Wilson’s area law

$$W(C) \simeq \exp(-\sigma \text{Area}(C)), \quad \sigma > 0$$

confirm the same fact in the infrared: a linear potential does not carry a massless vector. Therefore

$$\Delta = m > 0.$$

The gap belongs to A_G , not to the name “gluon.” Compact simple G survives as confined Yang–Mills. That arrangement has no massless wave.

The massless wave and the self-coupling are independent contributions. The commutation lemma says a linear map W superimposes independent additions:

$$W(x + c) = Wx + Wc.$$

In the logarithmic fold, multiplication is addition: independent contributions add in the fold and multiply in the answer. If x is the massless wave and c is the self-coupling, they must be kept as two terms. Written as one continuum Hamiltonian they smear a theory with $\Delta = 0$ across a theory with $\Delta > 0$. Inside A_G the massless value does not persist. Commutation is what forbids that smear.

$U(1)$ is the compact group with no self-coupling. Its surviving arrangement keeps the massless wave. Particle Data Group: $m_\gamma < 10^{-18}$ eV. That is $\Delta = 0$, which is why Jaffe and Witten wrote simple. Electroweak $SU(2) \times U(1)$ with a Higgs field is a different surviving arrangement. The W mass 80.36 GeV is the Higgs gate, not the glueball of pure compact simple G . Same engine rule: the gap belongs to the arrangement that survived.

Theorem 2. *For any compact simple G the massless infrared gluon does not persist. The first persisting knot has $m > 0$. That is a mass gap $\Delta > 0$.*

Proof. Simple $\Rightarrow A \wedge A$ is present $\Rightarrow$ a force mode sources itself $\Rightarrow$ free P_- at $m = 0$ is inconsistent. The first persisting mode is a P_+ glueball with $m > 0$. That infimum is Δ . Commutation keeps the $U(1)$ massless term off this arrangement, so it cannot cancel Δ . ■

6. Any compact simple G – their number Δ

Every compact simple G is nonabelian, so every such G has $A \wedge A$. The argument of Theorems 1 and 2 does not use which simple G . It uses compact and simple. Cost symmetry runs the same way

on each G : determined input, complete checker, unique-close. The number Δ can change with the rank of G . The sign of Δ cannot.

Jaffe and Witten named Δ as the first energy above the vacuum. Their construction never places that energy, because it starts from an undefined measure. This section places it on the exhibited field. Vacuum at 0. On Ψ_G ,

$$\mu_G = \frac{\int_{\Psi'' < 0} |\Psi''| d\sigma}{\int_{\Psi'' > 0} |\Psi''| d\sigma} = \sqrt{1 + 2\lambda_G}.$$

The energy unit is the string pin $\sqrt{\sigma}$. Then

$$\Delta = \mu_G \times \sqrt{\sigma}.$$

Compact simple G unique-closes Ψ_G , so both integrals are determined, so μ_G is determined. The lock identity $\lambda_G = \frac{\mu_G^2 - 1}{2}$ follows; it is not an inversion of a lattice mass. Lattice measures the same ratio as $m/\sqrt{\sigma}$. That is the check.

$\hbar c = 197.3269804$ MeV·fm. String pin $\sqrt{\sigma} = 440$ MeV. Confinement length $\hbar c / \sqrt{\sigma} = 0.4485$ fm.

group	lattice check	λ_G	Δ (MeV)	$\hbar c / \Delta$
SU(2)	Teper 3.74(12)	6.4938	1645.6(53)	0.1199
SU(3)	M-P 3.9318	7.2296	1730(50)(80)	0.1141
SU(3)	Teper 3.64(9)	6.1248	1601.6(40)	0.1232
SU(3)	Lucini-Teper 3.607(87)	6.0052	1587.1(38)	0.1243
U(1)	PDG $m_\gamma < 10^{-18}$ eV	—	0	none

SU(2) and SU(3) are two compact simple groups: same law, different pins, both $\Delta > 0$. $U(1)$ is compact and not simple: $\Delta = 0$. Electroweak $SU(2) \times U(1)$ with a Higgs field is a different arrangement; $m_W = 80.36$ GeV is not this glueball.

Those MeV values are Jaffe and Witten's Δ . Existence is still Theorem 1: the surviving arrangement. The table is that arrangement read in their units, on more than one compact simple G , which is what “any” requires. `python reproduce_ym.py` reprints every number.

Theorem 3. *Compact simple G implies existence on $\mathbb{R}^4$ and $\Delta > 0$. $\Delta = 0$ recovers $U(1)$, which is not simple.*

Proof. Forward: Theorem 1 plus Theorem 2. Back: $\Delta = 0$ means a massless knot persisted, so $A \wedge A$ was absent, so G was not simple. The compact group with that arrangement is $U(1)$. ■

7. The Gårding–Wightman theory

Section 1: Gårding–Wightman is a Hilbert space with a Poincaré action, a unique vacuum, and local field operators that commute at spacelike separation. Osterwalder–Schrader is the theorem that a family of Euclidean n -point functions with invariance, reflection positivity, and clustering determines such a Minkowski theory. Jaffe and Witten required existence at that strength. This section writes those n -point functions as output of the exhibited field A_G , checks the three hypotheses, and applies the theorem. It is the same theory as Theorems 1–3.

The Euclidean field is the exhibited 1-form $A_G = (P_+ + P_-)[\Psi_G]$ on $\mathbb{R}^4$, with curvature $F_G = dA_G + A_G \wedge A_G$. Let O range over gauge-invariant polynomials in F_G and its covariant derivatives, such as $\text{Tr } F_{ij}F_{kl}(x)$, which is what Jaffe and Witten required as local operators. Write $O(x)$ for that polynomial evaluated on this F_G at $x \in \mathbb{R}^4$. Compact G makes Haar measure finite, so these insertions are tempered distributions. They are evaluated on this field.

The vacuum Ω is the projection with no knot. Unique-close: exactly one consistent empty arrangement. Let the inner product on the positive-time half of the 4D slice be the Euclidean pairing. The chaos filter has already set every unstable mode to zero, so this inner product is positive. That positivity is reflection positivity. The Hilbert space $\mathcal{H}$ is the completion of persisting modes in that inner product.

The Euclidean n -point functions of this theory are the moments of those insertions in that vacuum:

$$S_n(x_1, \dots, x_n) = \langle \Omega, O_1(F_{G(x_1)}) \cdots O_n(F_{G(x_n)}) \Omega \rangle.$$

For test functions f_i in the Schwartz space $\mathcal{S}(\mathbb{R}^4)$,

$$S_n(f_1, \dots, f_n) = \int_{\mathbb{R}^4} d^4x_1 \cdots d^4x_n f_1(x_1) \cdots f_n(x_n) S_n(x_1, \dots, x_n).$$

Those are the correlators Osterwalder–Schrader takes as input. They are this pairing on the exhibited field A_G .

Euclidean invariance. The action S integrates d^4x over the whole slice. Euclidean isometries act by

$$(E(b, R)\Psi)(x, \sigma) = \Psi(R^{-1}(x - b), \sigma), \quad b \in \mathbb{R}^4, \quad R \in \text{SO}(4).$$

A_G is the unique survivor of an isometry-invariant action, so $E(b, R)$ preserves the vacuum and the insertions. Therefore S_n is translation-invariant and rotation-invariant.

Reflection positivity. The inner product on the positive-time half is the inner product of $\mathcal{H}$. Unstable modes are already gone. So the OS reflection-positivity inequality holds for these S_n .

Clustering. Theorem 2 and section 6: the first persisting knot has energy $\Delta = \mu_G \sqrt{\sigma} > 0$ for compact simple G , with μ_G the curvature ratio on Ψ_G . Therefore

$$|S_2(x, y) - S_1(x)S_1(y)| \leq C \exp(-\Delta |x - y|)$$

for large $|x - y|$, with this paper's Δ .

Osterwalder–Schrader reconstruction applied to this family $\{S_n\}$ yields a Gårding–Wightman theory on Minkowski $\mathbb{R}^4$: this $\mathcal{H}$; this Ω ; a unitary representation $U(a, \Lambda)$ obtained by continuing $E(b, R)$; and field operators $\Phi_O(f)$ whose Euclidean moments are these S_n . Explicitly

$$\Phi_O(f) = \int_{\mathbb{R}^4} d^4x f(x) O(F_{G(x)}).$$

Covariance. $U(a, \Lambda)\Phi_O(f)U(a, \Lambda)^{-1} = \Phi_O(f_{(a, \Lambda)})$, where $f_{(a, \Lambda)}(x) = f(\Lambda^{-1}(x - a))$.

Spectral condition. The chaos filter kills tachyons. Clustering at this Δ gives $\text{spec}(H) \subset \{0\} \cup [\Delta, \infty)$.

Locality. Spacelike-separated insertions are independent. The commutation lemma is $W(x + c) = Wx + Wc$. Therefore $[\Phi_O(f), \Phi_{O'}(g)] = 0$.

Cyclicity. Knots are gauge-invariant insertions on Ω . Unique-close generates $\mathcal{H}$ from Ω . So $\{\Phi_{O_1}(f_1) \cdots \Phi_{O_n}(f_n)\Omega\}$ is dense.

Short distance. Compact simple G has the Gross–Wilczek–Politzer β -function. At small $|x - y|$ the correlators S_n match perturbative Yang–Mills.

Gårding–Wightman	this construction
Hilbert space $\mathcal{H}$	persisting modes; inner product after the filter
vacuum Ω	no knot; unique-close
S_n	moments of $O(F_G)$
$\Phi_O(f)$	smearred $O(F_G)$
$U(a, \Lambda)$	continuation of $E(b, R)$
$\text{spec}(H) \subset \{0\} \cup [\Delta, \infty)$	clustering at $\Delta = \mu_G \sqrt{\sigma}$
locality	commutation lemma

Gårding–Wightman	this construction
cyclicity	knots from Ω
short distance	Gross–Wilczek–Politzer for simple G

Theorem 4. *The functions S_n constructed above by evaluating gauge-invariant polynomials on the exhibited curvature F_G of $A_G = (P_+ + P_-)[\Psi_G]$ are Euclidean-invariant, reflection-positive, and cluster at the rate Δ of section 6. Osterwalder–Schrader reconstruction therefore yields a Gårding–Wightman theory on Minkowski $\mathbb{R}^4$ whose Hamiltonian is this paper’s H and whose mass gap is this paper’s Δ . The field operators, the Poincaré action $U(a, \Lambda)$, and the spacelike commutators are the output of that reconstruction.*

Theorems 1 through 4 are the Clay sentence: for any compact simple G a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ as a Gårding–Wightman theory and has a mass gap $\Delta > 0$.

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Appendix — reprint

python reproduce_ym.py reprints the curvature-ratio lock, $\Delta = \mu\sqrt{\sigma}$, and the lattice check of μ .